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COA

CASE STUDY 5 :

{I/O Design for a Hospital Patient Monitoring Unit}

Introduction to the Problem :

In modern hospitals, patient monitoring units are used to continuously track vital parameters such as heart rate, oxygen level, body temperature. These sensors generate real-time data, which must be processed immediately to detect abnormalities and trigger alerts.

The system must :

- Continuously read sensor data.
- Respond quickly in emergency situations
- Ensure reliable and accurate data transfer
- Minimize processor overload.

Thus, designing an efficient input/output and data transfer mechanism is critical for patient safety.

Concepts from COA used in the solution

This problem involves several computer organization and architecture (COA) concepts :

- I/O organization
- Memory-mapped and I/O mapped addressing.
- Data transfer technique
- Interrupt Handling.
- Processor efficiency.

Memory - Mapped I/O

- I/O devices are treated as memory locations
- Same address space as main memory
- CPU uses normal instructions

Advantages
- easy programming
- No special instruction required
- supports large number of devices

Disadvantages → Reduces available memory space.

I/O - Mapped I/O

- separate address space for I/O devices
- special instruction are used.

Advantages → Does not affect memory space
→ clear separation of I/O and memory.

Disadvantages → limited address space
→ Requires special instructions

Comparison of Data transfer techniques:

(A) Programmed I/O

- CPU continuously checks device status
- Data transferred by CPU.

Advantages: simple to implement.

Disadvantage → wastes CPU time
→ Not suitable for real time system.

(B) Interrupt - Driven I/O

- Device sends interrupt when data is ready
- CPU responds only when needed.

Advantages → efficient CPU usage
→ faster response than polling.

Disadvantages → interrupt overhead
→ can become complex with many devices

ended Method for Real-time Patient Data.

for a hospital monitoring system, interrupt-Driven I/O combined with DMA is the best approach.

Justification:

- Patient data is continous and real time
- immediate response is required in emergencies
- CPU should not be overloaded

Working approach:

- sensors generate data → trigger interrupt
- CPU handles critical signals
- Large data transfers handled using DMA.

* Data Storage from input Devices to memory.

Data flow Process is

1. sensors collect patient data
2. Data is sent to I/O interface
3. Data is transferred to main memory
4. Data stored to buffer or memory location
5. CPU processes data for analysis

Sensors → I/O interface → (Interrupt / DMA) → Main memory → CPU → Alert system

- sensors acts as input devices
- I/O interface manages communication
- Memory stores data temporarily.
- CPU analyzes data and generates alerts

Effect on Processor efficiency :→

Programmed I/O :→ CPU busy all the time
→ very low efficiency

Interrupt-Driven I/O → CPU works only when needed
→ improved efficiency.

DMA → CPU free for other tasks
→ Highest efficiency

Proposed System Design :→

Hybrid I/O system : use interrupts for :→ emergency signals, Real time alerts
: use DMA for → Continous data transfer

Final Recommendation :→

The best solution → use memory mapped I/O for flexibility
→ use interrupt-Driven I/O for real time response
→ use DMA for efficient data transfer.

This combination ensures :→ ① fast response, ② High efficiency, ③ Reliable operati

Conclusion :→

Designing an efficient I/O system is crucial for hospital patient monitoring units by applying COA concepts such as I/O technique, memory mapping, data transfer method.
So we conclude that :→

- programmed I/O is not suitable due to inefficiency.
- Interrupt-driven I/O improves responsiveness
- DMA provides high speed data transfer.

A hybrid approach using interrupts and DMA offers the best performance for real-time medical applications, this ensures patient safety, system reliability, and optimal processor utilization

Working summary :→

① Sensors monitor patient
② Data sent via I/O interface
③ interrupt generated for critical
④ DMA transfers bulk data to memo
⑤ CPU processes and triggers alert

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COA

CASE STUDY 6 * {Storage planning for a Video Surrellance System}

* Introduction to the problem :-

In a campus security system, video surrellance is used to monitor activities continuously. Cameras operate 24x7 and generate a large amount of video data. This data must be stored efficiently and should be easily accessible when required for monitoring or investigation.

The major challenge in such system is to ~~desgi~~ design a storage solution that provides :

- High storage capacity.
- Fast data retrieval
- Reliability
- Cost efficiency.

To achieve this, the system must choose between magnetic disk storage (HDD) and flash - based storage (SSD) or use a combination of both.

* Concepts from COA used in the solution :

The following computer organization and Architecture (COA) concepts are applied :

- Memory Hierarchy : organizes memory into levels based on speed and cost.
  {Cache → RAM → Secondary Storage}
- Secondary Storage : Used for long - term storage of large data.
- I/O System : Handles continous input from cameras
- Performance Metrices : Access time, latency and throughput.
- Storage Trade-off : Balancing cost, speed and capacity.

These concepts help in designing an efficient and optimized storage system.

ROLE OF SECONDARY STORAGE :

Secondary storage is essential in a video surveillance system because :

- video data is very large and cannot be stored in primary memory.
- Data must be stored permanently for future use.
- Continuous recording requires high - capacity storage.

Functions :→

- Storing video recording
- Maintaining historical data.
- Supporting retrieval for playback.

Thus secondary storage acts as the backbone of the surveillance system.

Magnetic Disk vs Flash Memory.

⇒ Magnetic Disk {HDD}

- uses rotating magnetic platters.
- Data accessed using mechanical head.

Advantages : → Low cost per GB
→ High storage capacity.

Disadvantages : → Slow access time
→ Mechanical failure possible

⇒ Flash Memory {SSD}

- uses NAND flash memory.
- No moving parts.

Advantages : → High speed
→ Low latency
→ More reliable

Disadvantage : → Expensive
→ Limited write cycles

ole use of each storage type.

Both storage types are useful in different parts of the system:

- HDD is suitable for: Long term storage
  : Archiving old video.

- SDD is suitable for: Real-time recording
  : Frequently accessed data.

Practical usage :→

- SSD stores recent footage
- HDD stores older footage

Memory Hierarchy and Retrieval Speed.

memory hierarchy improves system performance by organizing storage based on speed :

1. Cache (fastest)
2. RAM
3. Secondary storage (SSD / HDD)

Impact:

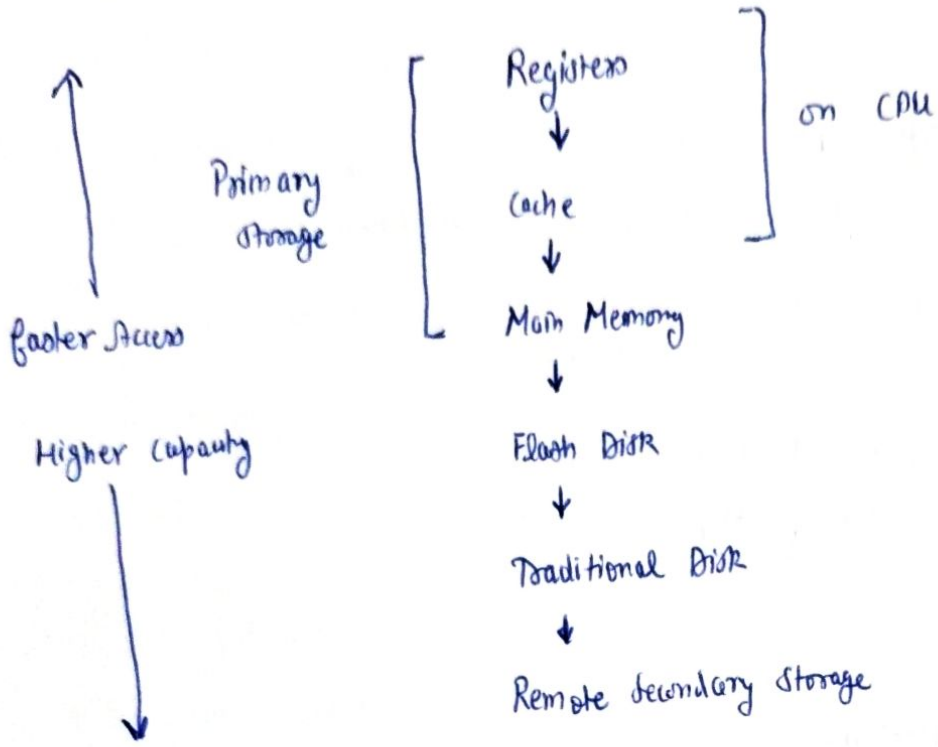
- Frequently accessed data stored in faster memory (SSD)
- Less frequently used data stored in HDD.

This reduces retrieval time and improves efficiency.

Proposed optimized Architecture :

Hybrid Storage System :

Working → 1. Cameras capture video
2. Data is stored in SSD
3. Frequently used data remains in SSD
4. Older data is moved to HDD
5. Retrival: • Recent data → SSD
   • Old Data → HDD.



The memory Hierarchy.

Analysis and Justification.

only HDD → Low cost but slow performance

only SSD → High speed but very expensive

Hybrid System

- Balanced cost and performance
- High capacity with faster access

this approach uses the strengths of both storage types effectively.

Final Recommendation :

The recommended solution is a hybrid storage architecture :

- use SSD for
  - Real time data
  - Frequently accessed footage
- use HDD for.
  - Long - term storage
  - Backup.